ANSI/IESLM-91-22

Illuminating
ENGINEERING SOCIETY

APPROVED METHOD: APPLICATION DISTANCE RADIOMETRY

AN AMERICAN NATIONAL STANDARD

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ANSI/IES LM-91-22

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Publication of this document
has been approved by IES.
Suggestions for revisions
should be directed to IES.

Prepared for IES
By the Testing Procedures Committee

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1.0 Introduction and Scope

1.1 Introduction

Certain luminaires are often used in near-field conditions, i.e., at working distances comparable to the largest light-emitting dimension of the luminaire itself. That can cause the use of far-field IES-format photometric files (the de facto default in all lighting, hereafter referred to as "IES files") to result in substantially incorrect irradiance, illuminance, or photon-flux density values being predicted by lighting layout models constructed for near-field conditions. This problem can be overcome by using distance-specific IES file(s), generated by first collecting data at the working distance(s) of interest obtained from a single luminaire and then converting them to distance-specific IES file(s), following the protocol described in this document.

In 2000, the IES Testing Procedures Committee developed the standard IES LM-70-00, Approved Guide to Near-Field Photometry. The intent of IES LM-70-00 (now deprecated) was to develop a test method to characterize intentional uplight. Indirect lighting, as it is commonly referred to, is often seen mounted 300 to 600 mm (approx. 12 to 24 inches) away from the intended plane of illumination, such as the ceiling. This document is intended to improve and expand the methods outlined in IES LM-70-00.

Photometric data collected in the method described in ANSI/IES LM-75-19 is commonly compiled in the data format defined in ANSI/IES LM-63-19. This data format and its application, in its current iteration, remain incompatible in near-field situations such as those required in UV germicidal and horticultural growth applications, among others. ANSI/IES TM-33-18 was developed specifically to support these requirements.

The IES Testing Procedures Committee developed this Approved Method for application-distance radiometry to address the near-field application of far-field photometric data obtained using the method described in ANSI/IES LM-75-19.

For the purpose of this document, the term *near field* applies to direct measurement at the range of an application distance that is less than five or ten times the longest dimension of the device under test (DUT).

Within this document, the term *irradiance* may be replaced with *illuminance* or *photon flux density,* as applicable, and the term *radiant flux* may be replaced with *luminous flux* or *photon flux*.

1.2 Scope

This document describes:

- The method for measuring illuminance, irradiance, and/or photon flux density at multiple points on a plane at a specific application distance
- A method to generate and interpret IES files composed of equivalent intensity values and applicable only to a specific range of application distances
- The method for merging far-field characteristics with near-field characteristics at angles where points on the plane may be in the far field

This document covers the testing of lamps, light engines, and luminaires. It covers light source technologies such as LED, incandescent, fluorescent, HID, and OLED sources. This document does not cover battery-operated light sources.

This document does not define methods for interpolating between measurement points on a given measurement plane, nor does it define interpolation methods between multiple measurement planes.

2.0 Normative References

This Standard is intended to be used in conjunction with the publications described in **Sections 2.1** through **2.9**; the latest edition of the publication shall apply.

Note: The technology-specific standards marked "conditionally normative" are normative depending on the technology of the DUT. For example, if a solid-state lighting product is being measured, then ANSI/IES LM-79 is normative but ANSI/ IES LM-41 and ANSI/IES LM-46 are not normative.

2.1 ANSI/ASABE S642 SEP2018
American Society of Agricultural and Biological Engineers. Recommended Methods for Measurement and Testing of LED Products for Plant Growth and Development. St. Joseph, Mich.: ASABE; 2018.

2.2 ANSI/IES LM-20-20
Illuminating Engineering Society. Approved Method: Photometry of Reflector Type Lamps. New York: IES; 2020.

2.3 ANSI/IES LM-41-20 (conditionally normative)
Illuminating Engineering Society. Approved Method: Photometric Testing of Indoor Fluorescent Luminaires. New York: IES; 2020.

2.4 ANSI/IES LM-46-20 (conditionally normative)
Illuminating Engineering Society. Approved Method: Photometric Testing of Indoor Luminaires Using High-Intensity Discharge or Incandescent Filament Lamps. New York: IES; 2020.

2.5 ANSI/IES LM-63-19
Illuminating Engineering Society. Approved Method: Standard File Format for the Electronic Transfer of Photometric Data and Related Information. New York: IES; 2019.

2.6 ANSI/IES LM-75-19
Illuminating Engineering Society. Approved Method: Guide to Goniometer Measurements and Types, and Photometric Coordinate Systems. New York: IES; 2019.

2.7 ANSI/IES LM-79-19 (conditionally normative)
Illuminating Engineering Society. Approved Method: Optical and Electrical Measurements of Solid-State Lighting Products. New York: IES; 2019.

2.8 ANSI/IES LS-1-21
Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2021. Online: https://www.ies.org/standards/definitions/. (Accessed 2022 Feb 1).

2.9 ANSI/IES TM-33-18
Illuminating Engineering Society. Standard File Format for the Electronic Transfer of Luminaire Optical Data. New York: IES; 2018.

3.0 Definitions

Definitions found in this section are for the purpose of this document. Definitions not found in this section may be found in *ANSI/IES LS-1-21, Lighting Science: Nomenclature and Definitions for Illuminating Engineering* (see **Section 2.8).**

3.1 beam angle
From ANSI/IES LS-1-21: The angle between the two directions for which the intensity is 50% of the maximum intensity as measured in a plane through the nominal beam centerline. For beams that do not possess rotational symmetry, the beam angle is generally given for two planes at 90 degrees, typically the maximum and minimum angles. Note that in certain fields of application, beam angle was formerly measured to 10% of maximum intensity.

In addition, for the purpose of this Approved Method: For beams where the intensity falls to 50% of the maximum more than twice within a given vertical plane, the beam angle shall be defined by the largest angles at which that happens. Beam angle is expressed in units of degrees.

3.2 device under test (DUT)
A device being measured according to the methods described in this document.

3.3 partial flux
The luminous, radiant, or photon flux passing through a defined area of an intended plane. These are expressed in units of lumens, watts, or micromoles per second, respectively.

If irradiance is the input, then partial radiant flux is the output. If illuminance is the input, then partial luminous flux is the output. If photon flux density is the input, then partial photon flux is the output.

4.0 Physical and Environmental Test Conditions

4.1 General

The requirements for the physical and environmental test conditions depend on the light source technology employed by the DUT. For example, some luminaires are sensitive to changes in temperature and air movement (see **Sections 4.2** and **4.3**).

4.2 Temperature

LED and fluorescent luminaires are often highly sensitive to changes in temperature and therefore require better temperature control in order for their results to reach lower levels of uncertainty. Incandescent and HID luminaires are not typically very sensitive to changes in temperature; however, laboratory temperature control is still required to ensure the proper performance of laboratory instruments. In all cases, the applicable requirements set forth in the following referenced standards, including measurement of ambient temperature, shall be met:

LED luminaires	ANSI/IES LM-79-19
Fluorescent luminaires	ANSI/IES LM-41-20
Incandescent and HID luminaires	ANSI/IES LM-46-20

It may be helpful to directly monitor the DUT temperature. (See **Section 10** for reporting requirements.) (For actively cooled DUTs, refer to ASABE S642 for more information; see **Section 2.1**.)

4.3 Airflow

LED and fluorescent luminaires are often highly sensitive to airflow, while Incandescent and HID luminaires are typically not affected. In all cases, the applicable requirements set forth in the following referenced standards shall be met:

LED luminaires	ANSI/IES LM-79-19
Fluorescent luminaires	ANSI/IES LM-41-20
Incandescent and HID luminaires	ANSI/IES LM-46-20

4.4 Humidity

High and low humidity can have adverse effects on test equipment and DUTs. The applicable requirements set forth in the following referenced standards should be met:

LED luminaires	ANSI/IES LM-79-19
Fluorescent luminaires	ANSI/IES LM-41-20
Incandescent and HID luminaires	ANSI/IES LM-46-20

4.5 Vibration

LED luminaires are generally unaffected by minor vibration; however, fluorescent, incandescent, and even HID luminaires can exhibit optical and electrical changes due to vibration. In all cases, good laboratory practice suggests that the DUT should not be subjected to excessive vibration or physical shock during stabilization, transportation, mounting, or testing.

4.6 Optical Conditions

4.6.1 Stray Light. Stray light should be suppressed in the test environment through the adequate use of low-reflectance finishes on surfaces, shielding, and baffling. The effectiveness of the stray light reduction in the laboratory can be evaluated by shielding the direct light of the source and measuring the stray light directly. (Refer to ANSI/IES LM-75-19 for more detailed information and requirements.)

4.6.2 Cosine Correction Requirements for Near-Field Measurements. (Refer to **Section 7.2**.)

4.6.3 Cosine Correction Requirements for Far-Field Measurements. The following applicable standards shall be followed:

LED luminaires	ANSI/IES LM-79-19
Fluorescent luminaires	ANSI/IES LM-41-20
Incandescent and HID luminaires	ANSI/IES LM-46-20

4.6.4 Wavelength Ranges. The measurement instrument should have sufficient wavelength range for the application. (See **Section 10** for reporting requirements.)

4.7 Spatial Conditions

4.7.1 Near-Field Scans.

4.7.1.1 Radial Method. The test system shall support a minimum spatial measurement range on the working plane of 1.75 m measured radially away from the vertical projection of the center of the DUT.

4.7.1.2 Rectangular Method. The test system should support a minimum spatial measurement range to characterize the DUT. The measured range or mirrored-data range shall be greater than the dimensions of the DUT's light emitting area. For example, if the DUT possesses 2-fold mirror symmetry and only one quadrant, with the vertical projection of the origin in one corner measured, then the measured range in the *x* direction shall be greater than ½ the DUT's *x* dimension, and the measured range in the *y* direction shall be greater than ½ the DUT's *y* dimension.

4.7.2 Far-Field Goniometry. The following applicable standards shall be followed:

LED luminaires	ANSI/IES LM-79-19
Fluorescent luminaires	ANSI/IES LM-41-20
Incandescent and HID luminaires	ANSI/IES LM-46-20

5.0 Electrical Test Conditions

5.1 Power Supply Requirements

Power supply requirements vary depending on the source technology used within the DUT. Different source technologies will challenge the power supply's ability to provide an electrical supply that is repeatable across laboratories while being representative of the performance of the DUT in a standard application. For example, the requirements for powering an LED DUT will be very different from the requirements for powering an incandescent or discharge-lamp DUT. In all cases, the applicable power supply requirements set forth in the following referenced standards shall be met (see **Section 10** for reporting requirements):

LED luminaires	ANSI/IES LM-79-19
Fluorescent luminaires	ANSI/IES LM-41-20
Incandescent and HID luminaires	ANSI/IES LM-46-20

5.2 Reference Circuit Requirements

No reference circuit is required for the scope of DUTs covered by this document.

5.3 Electrical Measurement Instrument Requirements

Measurement instrument requirements vary depending on the source technology used within the DUT. Some source technologies will challenge the measurement instruments in their ability to accurately measure the DUT during testing. For example, an LED DUT will likely challenge the electrical measurement instruments in different ways than an incandescent or discharge DUT will. In all cases, the applicable measurement instrument requirements set forth in the following referenced standards shall be met:

LED luminaires	ANSI/IES LM-79-19
Fluorescent luminaires	ANSI/IES LM-41-20
Incandescent and HID luminaires	ANSI/IES LM-46-20

(See **Section 10** for reporting requirements.)

6.0 Test Preparation

Before measurements are taken, the DUT shall be operated long enough to reach stabilization and temperature equilibrium. The method for stabilization and the time required depend on the light source technology employed by the DUT. In all cases, the applicable requirements set forth in the following referenced standards shall be met:

LED luminaires	ANSI/IES LM-79-19
Fluorescent luminaires	ANSI/IES LM-41-20
Incandescent and HID luminaires	ANSI/IES LM-46-20
Reflector type lamps	ANSI/IES LM-20-20

7.0 Test Procedures

Two methods of data collection are presented in this document: radial scan (using radial coordinates) and rectangular scan (using rectangular coordinates). The data collected from either method are not inter-convertible without a loss of accuracy. For either method, the alignment defined in **Section 7.1** shall be used.

The number of significant figures measured and reported should align with the application specific standards and industry practices.

7.1 Alignment and Coordinate System Establishment

The origin used in either method is defined as the geometric center of the DUT light-emitting area in the *x* and *y* dimensions and at the lowest *z* elevation of the DUT. The positive *z* direction is down, and the distance *z* is used in either method. The positive *x* and *y* directions follow the right-hand rule of Cartesian axis orientation, with the *y*-axis parallel to the longest edge of the emitting area (see **Figure 7-1**).

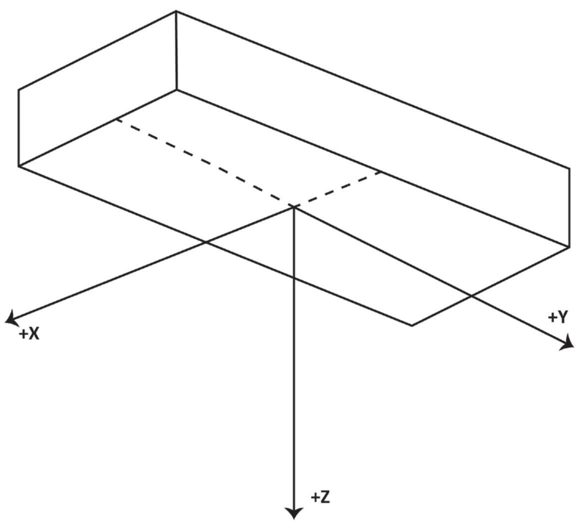


Figure 7-1. Coordinate system, including orientation of DUT.

In some applications, the light source's intended emission may be lateral or upward. In this case, the coordinate system should be rotated such that the positive *z* direction coincides with the intended emission direction and the above-mentioned criteria are maintained for the *x*-axis and *y*-axis.

7.2 Cosine Corrector Characterization for Near-Field Measurements

The cosine corrector device shall be characterized with respect to the cosine function. The error calculated in this section shall determine the minimum distance at which the DUT may be characterized. The cosine-corrector error characterization and error calculation shall be performed by the method described in this section.

A stable light source (not necessarily calibrated or traceable to SI units) may be used for this purpose in a controlled laboratory environment. The source needs to provide a uniform and consistent intensity in the solid angle subtended by the cosine corrector during the characterization. The light source shall be operated per the manufacturer's specifications. The light source shall emit in wavelengths that include the emission wavelengths of the intended DUT(s).

Stray light should be suppressed in the test environment, through the adequate use of low-reflectance finishes on surfaces, shielding, and baffling. In addition, stray light may be measured and subtracted from the detector response measurements (refer to ANSI/IES LM-75-19 for detailed information and requirements). Stray light measurements are made by placing a baffle or mask between the light source and the detector to block all direct rays of light, and then measuring through two opposite C-planes of measurement angles. The maximum stray light measurement after stray light reduction and/or correction (as applicable) shall not exceed 0.5% of the perpendicular (0°, unbaffled) measurement of the light source.

The procedure for cosine corrector characterization is described here in steps 1 through 7:

Step 1. Orient the cosine corrector such that it is facing directly at the light source (0° angle of incidence). The cosine corrector should be at a far-field distance where the sensor saturation is near the high end of the range recommended by the manufacturer of the spectroradiometer. This will utilize more of the dynamic range of the spectrometer and improve the signal-to-noise ratio at larger angles of incidence. The light should also be approximately collimated at the location where it interfaces with the cosine corrector. Optical collimation should be used if collimation cannot be achieved by spatially distancing the source from the cosine corrector. Given that either method of collimation can increase attenuation, the power level of the light source should also be considered.

Step 2. Confirm that the 0° signal was measured within 0.1° of the vertical axis through the geometric center of the cosine corrector, by using angle gauges as necessary. Make adjustments if needed.

Step 3. Once alignment is achieved, rotate the cosine corrector taking measurements between -85° and 85° in 5° increments.

Step 4. Return the cosine corrector to 0° and clock (i.e., rotate) the cosine corrector 90° about the optical axis.

Step 5. Repeat steps 2 and 3 to characterize the second plane. *Note:* As an alternative to steps 2 through 5, four perpendicular sweeps from 0° to 85° vertical may be performed such that the four 0° measurements can be evaluated to confirm proper alignment.

Step 6. Calculate the error for each of the four azimuthal angles as a function of zenith angle, $F(\theta)$:

$$F(\theta) = \left[\frac{E_e(\theta)}{E_e(0°)\cos(\theta)} - 1 \right] \text{, expressed as a percentage,}$$

where:
θ is the absolute angle of incidence
$E_e(\theta)$ is the irradiance as a function of angle of incidence

Step 7. Determine the minimum value F_{min} and the maximum value F_{max} of $F(\theta)$ among the four values obtained in Step 6 at each zenith angle, θ.

The tolerances allowed for F_{min} and F_{max} within each θ angle interval are specified in **Table 7-1**.

A detector shall not be used for distance-specific near-field IES file generation at or above the *lowest* θ value at which either F_{min} exceeds the minimal corresponding tolerance or F_{max} exceeds the maximal corresponding tolerance listed in **Table 7-1**. For any given detector, the minimal application distance shall be determined by:

$$z_{min} = 1.700/\tan(\theta_{high}) \text{ ,}$$

where z_{min} is the minimal acceptable application distance in meters and θ_{high} is the *highest* θ value at which the detector does not exceed the allowed tolerance on either side.

The detector may also be evaluated at θ intervals smaller than 5° below the initial θ_{high} value, in order to determine the latter more accurately and reach a shorter minimal application distance, if applicable. For example, if F_{max} exceeded the +10% tolerance limit at 85° in the original scan at 5° resolution (while F_{min} remained within its tolerance limits throughout), but a subsequent scan at 1° resolution found a θ_{high} value of 84° at which F_{max} still did *not* exceed +10%, then the corresponding z_{min} value is 0.180 m, rather than the original value of 0.302 m corresponding to the lowest θ angle of 80°, at which F_{max} and F_{min} are still within the tolerance limits at the original scan with 5° resolution in θ.

Table 7-1. Allowed Error Tolerances *F*(*θ*) for F_{min} and F_{max}

Vertical Angle Range*	Lower Error Limit	Upper Error Limit
(0,15]	-1.0%	1.0%
(15,20]	-1.5%	1.0%
(20,25]	-2.0%	1.5%
(25,30]	-2.5%	1.5%
(30,35]	-3.0%	2.0%
(35,40]	-3.5%	2.5%
(40,45]	-4.0%	3.0%
(45,50]	-5.0%	3.5%
(50,55]	-6.5%	4.0%
(55,60]	-8.0%	5.0%
(60,65]	-11.0%	6.0%
(65,70]	-14.0%	7.0%
(70,75]	-17.5%	8.0%
(75,80]	-21.0%	9.0%
(80,85]	-25.0%	10.0%

**Table note:* Vertical angles are shown as absolute values. A round bracket (parenthesis) indicates exclusion of the corresponding value, and a square bracket indicates inclusion.

7.3 Radial Scan Method

This method is specifically for generation of distance-specific IES files (ANSI/IES LM-63 or ANSI/IES TM-33).

Data shall be collected by measuring at specific near-field application distances with cylindrical coordinates and at far field distances with spherical coordinates. Beam angle may be determined by far-field data, and measurement resolution of the far-field scan may be as coarse as 5 degrees in θ. If the far-field beam angle varies around the z axis, then the smallest beam angle found shall be used for the measurements below:

For light sources with a far-field beam angle greater than or equal to 110°, the scan resolution shall have a maximum of 5° spacing in θ and a maximum of 22.5° spacing in ø. Finer spacing is always recommended for more accurate results in this and the following cases.

For light sources with a far-field beam angle *greater than or equal to 84*° but less than 110°, the scan resolution shall have a maximum of 3° spacing in θ and a maximum of 15° in ø.

For light sources with a far-field beam angle *greater than or equal to 58*° but less than 84°, the scan resolution shall have a maximum of 1° spacing in θ and a maximum of 9° in ø.

Light sources with a far-field beam angle less than *58*° should not be used for distance-specific IES file generation, due to the sharper increase in necessary resolution that is expected. For the same reason, application distances shorter than 15 cm should not be used for distance-specific IES file generation, regardless of the far-field beam angle of the light source.

7.3.1 Near Field – Direct Measurement at Application Distance. At near-field distances (see **Section 1.1 Introduction**), raw data shall be collected following the scan resolutions specified in **Section 7.3**, with the detector oriented along the vertical, until achieving either the largest θ value for the selected resolution that is still reachable within the minimal physical limit of a 1.75-m radial distance from the center of the DUT, or the largest θ value for which the cosine error of the detector calculated according to **Section 7.2** is still available, whichever occurs at the shortest radial distance. The θ value at the physical limit, θ_{lim}, can be calculated according to:

$$\theta_{\text{lim}} = \arctan\left(\frac{r_{\text{lim}}}{z}\right),$$

where:
r_{lim} is the actual physical distance at the physical limit for the test setup, in meters
z is the application distance in the vertical direction, in meters

For example, given r_{lim} = 1.75 m and an application distance of 0.305 m for a DUT with a far-field beam angle of 120° measured at a θ resolution of 3°, the θ_{lim} value is 80.1°, and the largest θ value required to collect direct measurements in near field would be 78°.

To correlate a radial distance, r, with the vertical angle, θ, this formula shall be used:

$$r = z \cdot \tan\theta$$

Any raw irradiance data values less than 0 shall be set to 0, to prevent modeling software crashes. **Figure 7-2** shows an example measurement web in the near field, and **Figure 7-3** shows an example measurement web in a single C-plane.

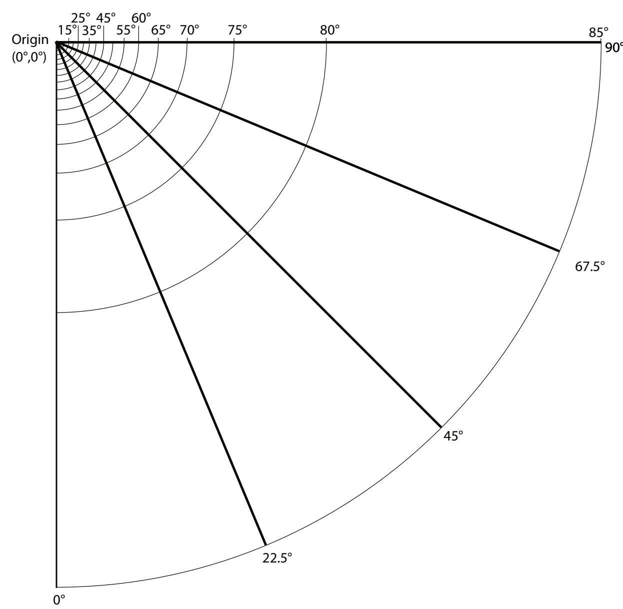


Figure 7-2. Type C measurement web projected onto the measurement plane, which is normal to the *z* axis. This depicts one-quarter of the measurement plane. This is sufficient when there are two mirror planes of symmetry passing through the C0 and C90 planes.

The origin of the measurement coordinate system shall be chosen on the DUT side facing the working plane, with the positive z direction always pointing toward the working plane. If the origin of the coordinate system thus chosen, or any other point on the light emitting surface of the DUT facing the working plane, is obscured by the base of the DUT (for example, in a table lamp or a pendant luminaire), the base may be removed for this

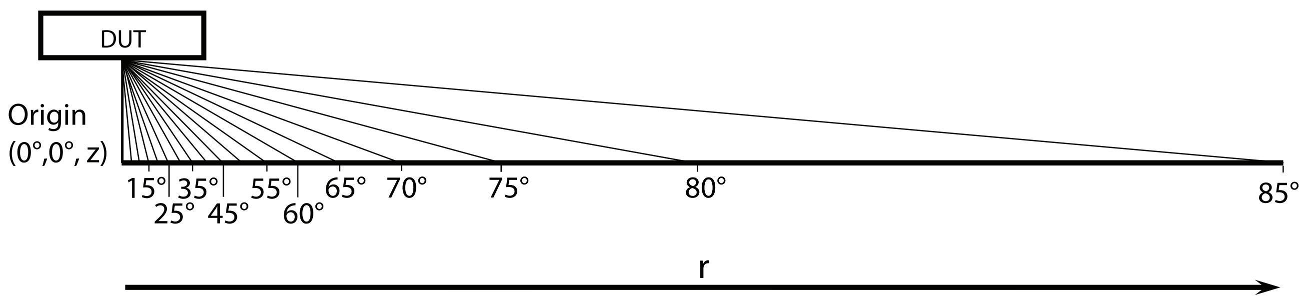


Figure 7-3. Side view of the measurement web under the DUT.

particular measurement. (See **Section 10** for reporting requirements.)

During this measurement, the DUT should be oriented in its intended operating position with respect to gravity. If the detector setup cannot accommodate measuring the DUT in such a position (for example, if the detector is permanently oriented upward and the DUT is a chandelier, with the ceiling being its working plane), the DUT may be operated in a different position, allowing this particular measurement. (See **Section 10** for reporting requirements.)

Note: The flux emitted by the DUT in this measurement is corrected for the intended operating position by the method described in **Section 9.2**. However, that correction does not reverse any changes in the light distribution pattern or the spectral power distribution resulting from unintended operation positions. If such potential changes are of major concern to the end user, they can be investigated additionally in customized measurements outside the scope of this document.

7.3.2 Far Field. Far-field measurements shall be performed per the following conditionally normative standards, depending on the light source technology employed by the DUT:

LED luminaires	ANSI/IES LM-79-19
Fluorescent luminaires	ANSI/IES LM-41-20
Incandescent and HID luminaires	ANSI/IES LM-46-20
Reflector type lamps	ANSI/IES LM-20-20

If any of the additional requirements listed further below in this section contradicts a requirement from an applicable conditionally normative standard, the requirements in this section shall prevail for this measurement.

In all cases, the coordinate system depicted in **Figure 7-4** shall be used and shall be oriented the same way as described in **Section 7.3.1**. The DUT itself shall be oriented in its intended operational position with respect to gravity for this measurement. (See **Section 10** for reporting requirements.)

If the far-field measurement system software cannot accommodate the coordinate system orientation required in this section, the far-field data may be first acquired as the software allows and then they may be re-oriented numerically by coordinate transformations as needed. (For additional information on numerically re-orienting the coordinates, see **Annex B**.)

The vertical and horizontal angular resolutions shall match those of the near-field measurements taken according to **Section 7.3.1** (whereas the resolution of the latter measurements in the vertical direction may be lower in comparison), with the following exception:

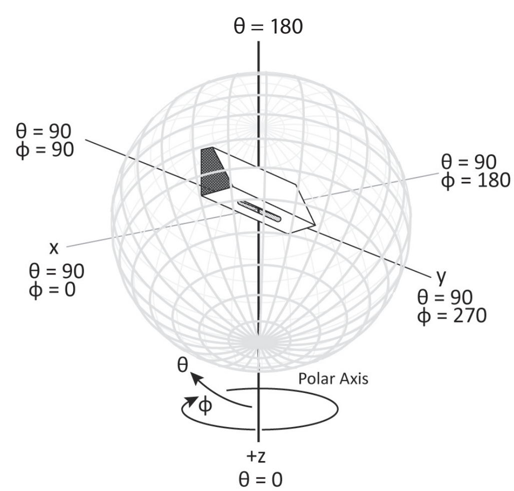


Figure 7-4. Coordinate system used in far-field measurements.

In the case of vertical rotational symmetry of the DUT beam in far-field measurements, a single horizontal angle value of 0° may be used for the far-field IES file (see also **Section 9.4** and the examples given in **Annex B**.) In all cases, the remaining applicable requirements set forth in the above referenced standards shall be met.

7.4 Rectangular Scan Method

For certain purposes, such as simple graph generation, data may be collected with Cartesian coordinates (see **Figure 7-5**). The rectangular scan method shall not be used to generate files in intensity units. This section describes the method for determining performance characteristics of products intended for application-distance-specific purposes. Performance characteristics include irradiance spatial distribution on an intended plane; optionally, partial radiant flux at the intended plane may be reported (see **Section 10.6**). By default, SI units should be used. However, other units may be used as requested by the initiator of the test.

If it is assumed that the emission of the DUT will demonstrate 1-fold or 2-fold mirror symmetry, then a half-plane or quarter-plane, respectively, may be captured and mirrored for reporting purposes. (See **Section 10** for reporting requirements.)

Once the measurement grid is defined, the characteristic of choice (i.e., illuminance, irradiance, or photon flux density) shall be measured at each location on the grid. Any data values less than 0 shall be set to 0 for display purposes.

7.4.1 System and Sensor Setup.

7.4.1.1 Equipment Description. A motorized XYZ gantry is arranged such that the x- and y-axes are level with respect to gravity. A cosine-corrected sensor is installed on the z-axis such that the measurement aperture is parallel to the ground and directed upward.

7.4.1.2 DUT Setup. The DUT should be immobilized using the manufacturer's recommended mounting method. The DUT should be oriented such that its emitting surface is level with respect to gravity and emitting downward. The DUT should be aligned such

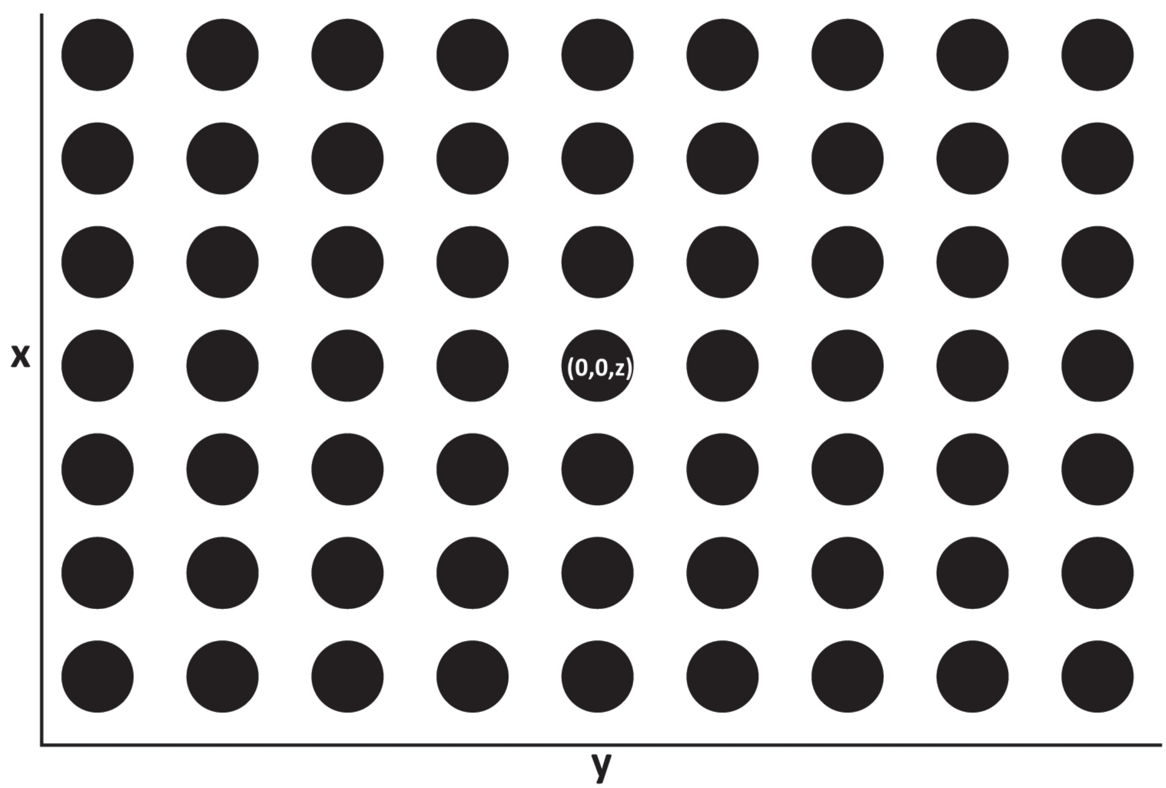


Figure 7-5. Measurement grid and coordinate system for rectangular scan method.

that the longest edge of the DUT emitting area is parallel to the y-axis of the gantry system.

7.4.1.3 Measurement Points. The distance between any two adjacent measurement points in either the x or the y direction should not exceed 150 mm. For devices with narrow emission angles or non-uniform emission, closer spacing of measurements is recommended.

The overall size of the measurement plane may be specified by the manufacturer. It is recommended that this be greater than 1,200 mm in both the x and y dimensions.

Measurements should be spaced consistently in the x direction and consistently in the y direction.

The z value (i.e., distance from emitting surface to measurement plane) should be set according to test requester's specifications. The limit of z_{min} as defined in **Section 7.2** applies to the rectangular scan. This method should not be used for devices with beam angles less than 60 degrees.

In the cases of high-power devices measured at small distances or for long test durations, it is recommended that multiple dark measurements be taken, especially when measuring with sensors that are not temperature stabilized.

8.0 Data Processing

This section describes the method for calculating partial flux (refer to **Section 3.3**) from data obtained using the rectangular scan method (see **Section 7.4**). (*Note:* When comparing these results to modelled data, some commercially available modelling software does not use corner and edge points by default, but they may be added manually.)

Partial radiant flux through the finite measurement plane may be calculated according to one of the two equivalent methods described in **Section 8.1 Four Corners Method** or **Section 8.2 Weighted-Average Method.**

8.1 Four-Corners Method

Four adjacent measurement points form a rectangle and a uniform irradiance value is assigned to the area of that rectangle which is equal to the average of the four measurements. The total flux through that rectangle is equal to the average irradiance times the area of the rectangle. Then the flux of each square is summed to determine the flux through the entire measurement plate.

For a rectangular grid consisting of N by M points (see **Figure 8-1**):

$$\Phi = A_{\mathrm{r}} \sum_{j=1}^{(N-1)(M-1)} \overline{E}_j ,$$

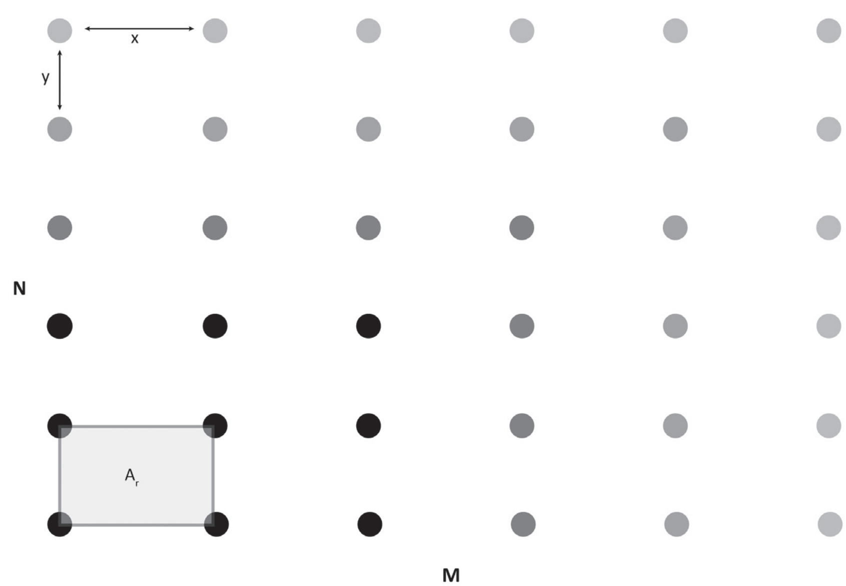


Figure 8-1. Depiction of four-corners method.

where:

Φ is the total flux through the measurement plane

A_r is the area of a single rectangle

j is the number of rectangles

$\overline{E}$ is the averaged irradiance of the four corners of the rectangle

8.2 Weighted-Average Method

The measured irradiance value can be assigned to a rectangle with the measurement point at its center. However, for edge measurements, the area shall be halved, and for corner measurements, the area shall be quartered. Interior measurements are weighted with a factor of 1.

For a rectangular grid consisting of N by M points (see **Figure 8-2**),

$$\Phi = A_r\left(\tfrac{1}{4}\sum_{c=1}^{4} E_c + \tfrac{1}{2}\sum_{e=1}^{2(N-2)+2(M-2)} E_e + \sum_{i=1}^{(N-2)(M-2)} E_i\right),$$

where:

Φ is the total flux through the measurement plane

A_r is the area of a single rectangle

c is the number of corner rectangles

e is the number of edge rectangles

i is the number of interior rectangles

E is the irradiance measured at each corresponding point

9.0 Procedures for Application-Distance IES File Generation

The steps described in **Sections 9.1** through **9.4** shall be followed.

9.1 Initial Data Transformation

The initial irradiance data collected and the application plane according to **Section 7.3.1** shall be transformed according to the following formula, to calculate the raw equivalent radiant intensity:

$$I_{raw} = Ez^2\cos^{-3}\theta,$$

where:

I_{raw} = raw equivalent intensity

E = experimental data, obtained with the detector aligned perpendicularly to the working plane

z = vertical distance from the optical center of the luminaire to the working plane

θ = angle, measured from the geometric vertical

The resulting raw equivalent intensities shall be corrected, as described in **Section 9.2**.

9.2 Near-Field Data Correction

This correction shall be performed on all near-field intensity values obtained according to **Section 9.1 Initial Data Transformation**. The correction factor shall be calculated according to the formula:

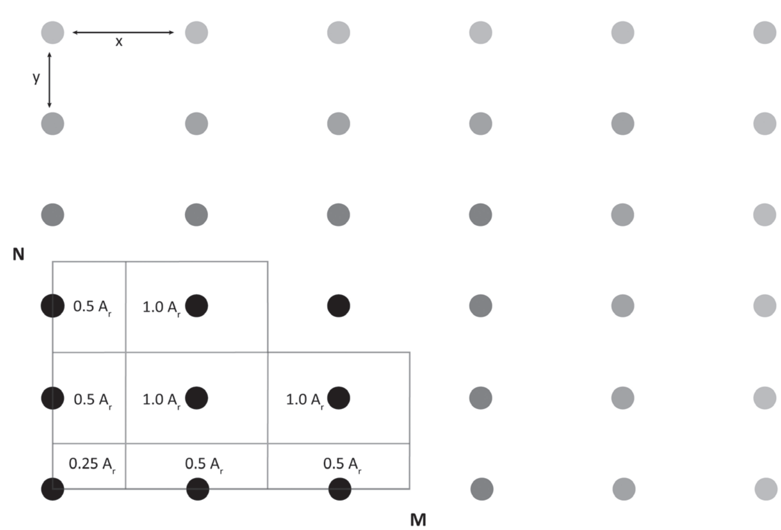


Figure 8-2. Depiction of the weighted-average method.

$$C = \frac{\Phi(\text{far field, 0 to } \theta_{\max})}{\Phi(\text{near field, 0 to } \theta_{\max})} \ ,$$

where:

C = the correction factor used to multiply each corrected intensity value prior to its inclusion in the application distance IES file

$\theta_{\max}$ = the maximal vertical angle at which data were collected in the application plane (the smaller of the θ_{high} value [see **Section 7.2**] and the θ_{lim} value [see **Section 7.3.1**])

Φ(far field, 0 to $\theta_{\max}$) = radiant flux calculated from far field data from $\theta = 0$ out to $\theta = \theta_{\max}$, where the θ values from 0 to $\theta_{\max}$ of the far-field data match those of the near-field data, per the requirements of **Section 7.3.2**. Far-field data collected beyond $\theta_{\max}$ will be used differently, as described in **Section 9.3**.

Φ(near field, 0 to $\theta_{\max}$) = radiant flux calculated from the near field I_{corr} data out to $\theta_{\max}$;

$I_{\text{corr}} = C{\cdot}I_{\text{raw}}$

The flux values used for the correction factor above can be calculated by numerical integration:

$$\Phi = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\theta_{\max}} I(\theta, \phi) \sin\theta \, d\theta d\phi \ ,$$

where the values of θ and ø are in radians.

The same numerical integration method shall be used for far-field and near-field data measured with any given DUT according to the methods described in this standard. Different modeling software products may use different integration methods (e.g., rectangle, Simpson's). Those are acceptable for all practical applications as long as they provide near-field flux values corrected to within 0.5% of those for the far-field flux.

9.3 Merging Data at Transition to a Vertical Angle of 90° and Beyond

Any intensity values at vertical angles below 90° for which no raw data could be measured on the near-field working plane due to a $\theta_{\max}$ value below 90° shall be substituted with the "conventional" intensity values from the far-field IES file, obtained per ANSI/IES LM-75 and oriented according to the conditions described in **Section 7.3.2**. Those far-field intensity values shall be merged directly with the near-field data into a single distance-specific IES file, without being subjected to the correction described in **Section 9.2**. (See **Annex B** for an example.)

Light intensity data for $\theta \geq 90°$ may be added into a near field IES file directly from the far field intensity data (indexed as directed in **Section 7.3.2** and without any further corrections) for modeling the DUT in the far field away from the working plane. This is particularly relevant to bidirectional and omnidirectional DUTs.

If no light emission at $\theta \geq 90°$ is expected or considered from the DUT, all intensity values corresponding to that θ range may be assumed to be 0 for the purpose of near-field IES file generation. The value of 0 may also be added explicitly into the distance-specific IES file for this case, as needed.

9.4 Application Distance IES File Generation

The application distance (near-field) IES files generated according to this procedure are usable with modeling software in a way similar to that for far-field IES files, with the following key differences:

- No near-field application distances other than the one listed for the near-field IES file shall be used. There is no mechanism built into either the ANSI/IES TM-33-18 or the ANSI/IES LM-63-19 file format to prevent modeling software from applying near-field IES files to any working distance, so setting the correct near-field distance in the model calculations is the responsibility of the user in each case. However, such files may be used for far-field modeling at 90° or more from the vertical to the near-field operation plane if containing such far-field data added according to **Section 9.3**.
- All luminous dimensions of the luminaire shall be set either to 0 (for modeling software accepting 0 as a luminous dimension) or to the smallest physical value acceptable by modeling software that will not accept luminous dimensions of 0, in the units specified in the distance-specific IES file. For example, luminous dimensions of 0.001 m or ft would be acceptable in DIALux rather than values of 0, where the latter would cause a default to much

larger values with that software. Using the actual luminous dimensions of a luminaire will generally give incorrect results with near-field IES files, due to misinterpretation of the latter's mathematical properties by the software. (Refer to **Section A.2** in **Annex A** for additional information.) If the modeling software also accepts geometric dimensions of the luminaire different from its luminous dimensions with no impact on lighting calculations, then the geometric dimensions may be added by hand during the model setup, for drawing purposes only.

- Application distance IES files containing a single horizontal angle of 0° shall not be generated. However, far-field IES files containing a single horizontal angle (if justified by luminaire beam symmetry) may be used for the purposes of the merging procedure described in **Section 9.3**, added with the same values for the same vertical angles (θ) at multiple horizontal angles (ϕ). Application distance IES files collected to horizontal angles of 90°, 180°, or 360° are acceptable if they follow the symmetry of the near-field intensity data, which may be a lower order of symmetry than that of the corresponding far-field data for any given DUT. (For the purposes of this document, rotational symmetry around the z axis, meaning lateral symmetry in all planes, is assumed to be the highest order of symmetry; the presence of two perpendicular mirror planes containing the z axis (meaning lateral symmetry in each quadrant) is the second highest order of symmetry; the presence of a single mirror plane containing the z axis, meaning lateral symmetry in only two half-planes, is the third highest order of symmetry; and the absence of all preceding symmetry elements is the lowest order of symmetry.) For example, if the far-field dataset has no axial symmetry but has two mutually perpendicular mirror symmetry planes containing the z axis, such a far-field dataset, with ϕ = 90°, would be acceptable to use with near-field data from the same DUT where ϕ = 90°, 180°, or 360°, provided the latter are collected at the same θ and ϕ increments as for the far-field data.
- The near-field IES file may need to be re-oriented with respect to gravity within the software model in certain cases, which may not require the same to be done for far-field IES files—for example, a pendant luminaire with the ceiling of the room as its working plane. (Luminaire re-orientation is trivial for modeling software.)

Application distance IES files can be written in the format specified by ANSI/IES TM-33-18 or ANSI/IES LM-63-19, depending on which is supported by the modeling software of the end user. (See **Annex B** for examples given in both IES file formats.) Additional software performing file conversion between these two formats is available. Other functionally equivalent file formats may be used if required by the modeling software.

An application distance IES file shall contain the following additional information:

1) A clear indication that it is meant only for near-field calculations at a specified single distance from the luminaire; this information shall be given *directly below* the test name, number or description.

   In the ANSI/IES LM-63-19 format, this shall appear thus:

   ```
   [TEST] (test name, number or description)
   [NEARFIELD] (application distance in m),0,0
   [MORE] THIS FILE MAY NOT BE USED FOR ANY APPLICATION DISTANCES OTHER THAN THE ONE LISTED ABOVE.
   ```

   In the ANSI/IES TM-33-18 format, this shall appear thus:

   ```
   <Comment>[TEST] (test name, number or description) </Comment>
   <Comment>[NEARFIELD] (application distance in m),0,0</Comment>
   <Comment>[MORE] THIS FILE MAY NOT BE USED FOR ANY APPLICATION DISTANCES OTHER THAN THE ONE LISTED ABOVE.</Comment>
   ```

2) A warning against using the actual luminous dimensions of the luminaire for near-field calculations, *optionally* followed by the physical dimensions of the

luminaire for drawing purposes and by any additional information about the shape of the luminaire if needed.

In the ANSI/IES LM-63-19 format, this shall appear thus:

```
[OTHER] THE ACTUAL LUMINOUS DIMENSIONS MAY NOT BE USED IN NEAR-FIELD LIGHTING CALCULATIONS.
[OTHER] LUMINAIRE PHYSICAL DIMENSIONS: (width, length, height in m) m
[OTHER] (any additional information describing the luminaire, such as the shape of the luminaire)
```

In the ANSI/IES TM-33-18 format, this shall appear thus:

```
<Comment>[OTHER] THE ACTUAL LUMINOUS DIMENSIONS MAY NOT BE USED IN NEAR-FIELD LIGHTING CALCULATIONS.</Comment>
<Comment>[OTHER] LUMINAIRE PHYSICAL DIMENSIONS: (width, length, height in m) m</Comment>
<Comment>[OTHER] (any additional information describing the luminaire, such as the shape of the luminaire)</Comment>
```

If the detector is limited by its vertical angle error, the minimal application distance z_{min} at which it may be used (calculated per **Section 7.2**) shall also be listed. In the ANSI/IES LM-63-19 format, this shall appear thus:

```
[OTHER] MINIMAL APPLICATION DISTANCE ALLOWED BY THE DETECTOR ERROR: (distance in meters) m
```

In the TM-33-18 format, this shall appear thus:

```
<Comment>[OTHER] MINIMAL APPLICATION DISTANCE ALLOWED BY THE DETECTOR ERROR: (distance in meters) m</Comment>
```

If far-field data are being merged into the application distance file, the first angle at which this is done shall also be listed. In the ANSI/IES LM-63-19 format, this shall appear thus:

```
[OTHER] VERTICAL ANGLE WHERE NEARFIELD TRANSITIONS TO FAR-FIELD: (angle in degrees) deg
```

In the ANSI/IES TM-33-18 format, this shall appear thus:

```
<Comment>[OTHER] VERTICAL ANGLE WHERE NEARFIELD TRANSITIONS TO FARFIELD: (angle in degrees) deg</Comment>
```

Any symmetry assumption shall be reported as described in **Section 10** (see also examples in **Annex B**).

In the ANSI/IES LM-63-19 format, this shall appear thus:

```
[OTHER] ASSUMED SYMMETRY: (specify symmetry)
```

In the ANSI/IES TM-33-18 format, this shall appear thus:

```
<Comment>[OTHER] ASSUMED SYMMETRY: (specify symmetry)</Comment>
```

10.0 Reporting Requirements

In addition to the reporting requirements defined in the technology-specific normative documents (ANSI/IES LM-79, ANSI/IES LM-41, ANSI/IES LM-46, and ANSI/ASABE S642; see **Section 2**), the information described in **Sections 10.1** through **10.5** shall be reported.

10.1 Administrative Information

a) Testing agency identification

b) Testing and report date

10.2 DUT Description

a) DUT manufacturer's name

b) Description of DUT, including model number

10.3 Test Conditions

a) Ambient conditions as may be required by the technology-specific standards (refer to **Section 4.2**)

b) DUT temperature and measurement location, if measured

c) Any removal of the DUT base for near-field measurement, per **Section 7.3.1**

d) Any change from the intended operation position of the DUT with respect to gravity in near-field measurement, per **Section 7.3.1**

10.4 Test Equipment

a) Short description of testing equipment, if applicable

b) Wavelength range of the optical measurement

10.5 Measured Results

a) Electrical input parameters

b) Coordinate system used

c) Any symmetry assumed

d) Application distance(s) used for measurements

e) Measurement points

f) Measured optical data (as applicable):

- Flux (or partial flux, if calculated)
- Intensity (may be submitted in ANSI/IES LM-63 or ANSI/IES TM-33 file format)
- Irradiance (for rectangular scan)
- Spectral power distribution (SPD)
- Wavelength range (for radiant and quantum measurements)

10.6 Optional Items

Any additional information required by the test requestor or regulatory body.

Annex A – Exact Near-Field Models and Related Insights

This annex summarizes the extensive mathematical analysis using optical models, performed by Technical Group C-303 of the IES Testing Procedures Committee. Special care was taken to maximize model accuracy and to incorporate the experimentally measured detector error, as described in the sections that follow. All recommendations by the group regarding vertical and horizontal angular scanning resolutions in the data collection step are based on this analysis, as are all procedures for near-field data corrections and extension at high vertical angles provided in the main document.

A.1. Setting Up the Initial Optical Models

The following ten steps were taken:

Step 1. A primary light source was defined (a mid-power LED with a square light-emitting surface 3 mm x 3 mm in size).

Step 2. For best reproducibility of this analysis, IES files were generated mathematically for five different beam profiles using the analytical formulas provided in **Table A-1** (following Step 10). Each primary light source was assumed to provide Φ = 100 lm, and no further beam modifications by secondary optics were added, for simplicity's sake.

Step 3. A model assembly of the primary sources was created, forming the basic luminaire to be modeled (2 rows of 50 each square 3030 LEDs, spaced 2 cm apart in both (*x* and *y*) directions). The typical far-field requirement (10 times the largest LED dimension) for the LED source described here is satisfied at a working distance of 4.3 cm (1.7 in.).

Important: Any optical model calculated for a working distance beyond that can therefore be considered *exact* in terms of satisfying the near-field requirement at luminaire level. This is what is meant by the use of the expression “exact model(s)” throughout this annex.

Step 4. Optical models were set up using DIALux v4 software. Unlike ray-tracing software, DIALux doesn’t use the Monte Carlo method, so it gives precisely the same outputs from the same inputs every time. This lack of random computational noise from the software was very useful for the exact models calculated by the Technical Group.

Step 5. The luminaire assembly was modeled in DIALux at different application distances, *each one satisfying the far field requirement for the primary source.*

Step 6. The illuminance distributions on the resulting horizontal calculation planes were examined and compared.

Step 7. Horizontal calculation areas of 1.500 m x 1.500 m were used, with special attention to those at 15 cm (approx. 6 in.) and 30 cm (approx. 12 in.) from the luminaire. High-resolution illuminance maps and 9 x 9 point calculation grids (excluding the edges) were obtained for statistical analysis. A DIALux setup over a calculation plane with the specified calculation grid is shown in **Figure A-1**, viewed from above. The square LEDs in the luminaire are visible as two rows of dots over the calculation grid points in the center of the grid.

Step 8. Both near-field and far-field models (the latter using luminaire-level theoretical IES files with Φ = 10 kilolumens [klm]) and the physical outline of the luminaire-level luminous surface) were calculated and compared.

Step 9. The initial analysis used 19 vertical angles at every 5° and 16 "C" planes at every 22.5°. Subsequently, resolution was increased to 1° for vertical angles and 3° for horizontal, for a total of 120 "C" planes. Intermediate resolution cases were also examined, in order to make appropriate final recommendations to the end users.

Step 10. Five different theoretical beam profiles were used, with beam angles ranging from 60° to 150°, based on analytical formulas defining them as functions of the vertical angle θ, with nicknames assigned for easier reference, as shown in **Figure A-2** and **Table A-1**.

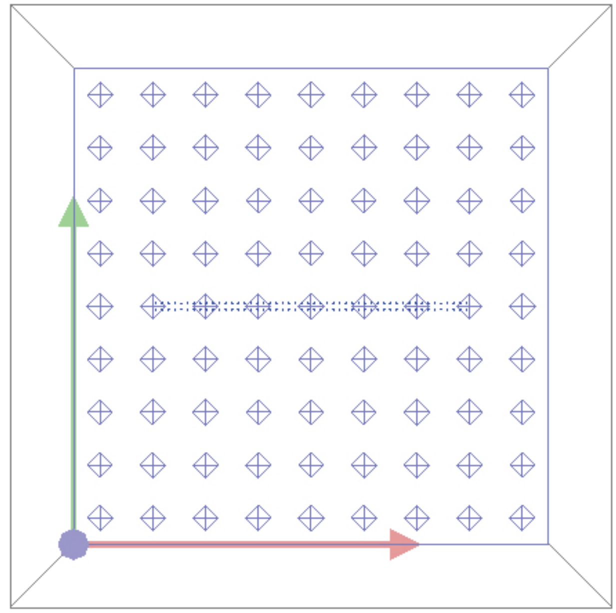


Figure A-1. Calculation grid (represented by the diamond shapes) used for statistical analysis, shown over the calculation area. (Graphic courtesy of Emil Radkov)

In **Table A-1**, Φ is the radiometric flux of either the primary light emitter (for near-field models) or the entire luminaire (for far-field models). The beam angle of the "bumblebee" luminaire was defined using the

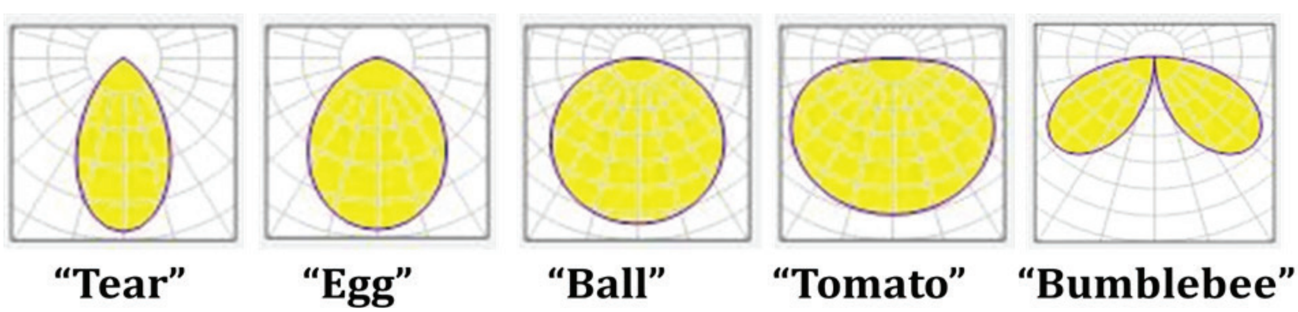


Figure A-2. Polar diagrams and nicknames used for the five theoretical beam profiles. (Graphic courtesy of Emil Radkov)

Table A-1. Beam Angles and Analytical Formulas Used to Generate the Five Theoretical Beam Profiles

Nickname	"Tear"	"Egg"	"Ball"	"Tomato"	"Bumblebee"
Beam Angle	Approx. 60°	90°	120°	Approx. 150°	150°
Formula	$\Phi\left(\frac{3}{\pi}\right)\cos^5\theta$	$\Phi\left(\frac{3}{2\pi}\right)\cos^2\theta$	$\Phi\left(\frac{1}{\pi}\right)\cos\theta$	$\Phi\left(\frac{3}{4\pi}\right)\cos^{1/2}\theta$	$\Phi\left(\frac{3}{2\pi}\right)\sin\theta\cos\theta$

broadest vertical angles (in this case, ±75°) at which that beam's intensity falls to half the maximum level. (Refer to **Section A.6** for justification of this definition.)

A.2. Generation and Initial Analysis of Near-Field IES Files

The steps listed below were followed:

Step 1. Exact models were used to calculate illuminance values in x-y planes on radial grids, subsequently scaled by $r^2 \cos^{-1}\theta$ (i.e., $z^2 \cos^{-3}\theta$) to generate near-field (NF) IES files and DIALux models at z = 0.150 m and z = 0.300 m. An example of one such radial grid (used at a working distance of 0.300 m and in 5° vertical angle increments) is shown in **Figure A-3**.

Step 2. The NF IES files generated in Step 1 were compared in polar plots, zonal lumens, and total lumens to their far-field (FF) counterparts, constructed at luminaire level from the same theoretical formulas.

Step 3. Illuminance maps and descriptive statistics (minimum, average, and maximum values; and root-mean-square errors (RMSE) vs. the exact models for 9x9 calculation grids at working distances matching those of the corresponding NF IES files) were also calculated and analyzed.

The following bullet items provide a summary of the salient differences observed between the FF and NF IES files of this case study:

- Regardless of the original FF beam shape, all NF IES files had "wings" resulting from the luminaire's length. The polar diagrams in **Figure A-4** show an example for the IES files of the "ball" beam shape.

  Note: The "ball" (better known as Lambertian) beam profile is a good approximation for luminaires using LEDs with flat tops and no further beam shape modification, which are common nowadays.

- For each beam type, the vertical angle of the NF "wings" increased as the working distance decreased.

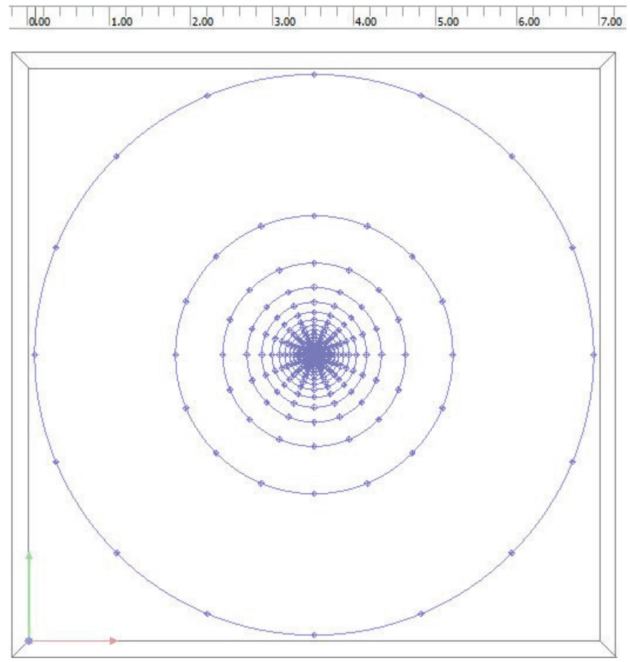


Figure A-3. Full radial calculation grid used for initial analysis. (Graphic courtesy of Emil Radkov)

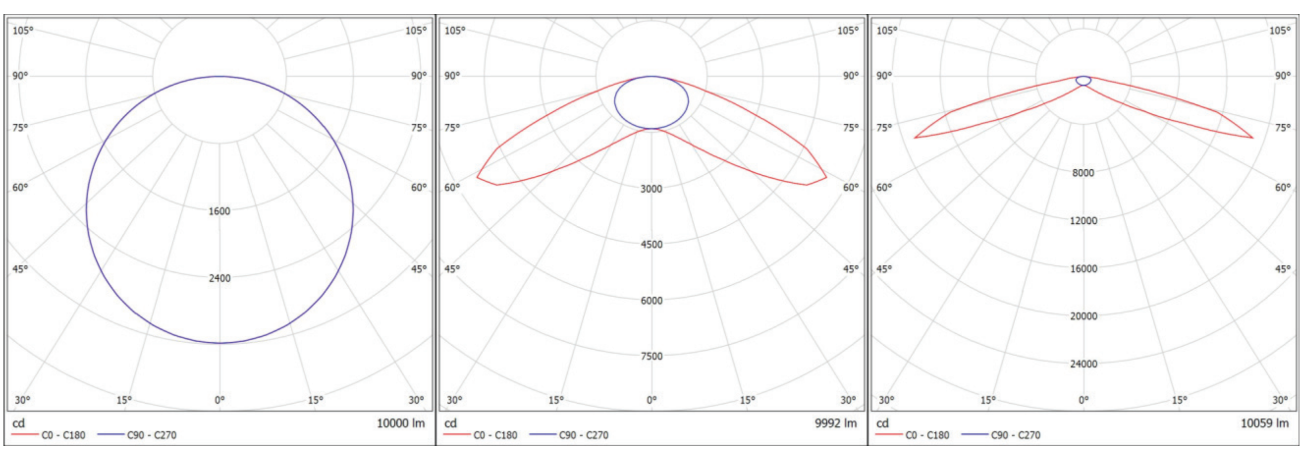


Figure A-4. Polar diagrams of IES files for the "ball" beam in far field (left), at 0.300 m near field (middle), and at 0.150 m near field (right). (Graphics courtesy of Emil Radkov)

- For each beam type, the width of each NF "wing" decreased with working distance.
- The total lumen deviations from FF calculated for the NF files generally increased as the beam angle decreased, and always more so at shorter working distances.

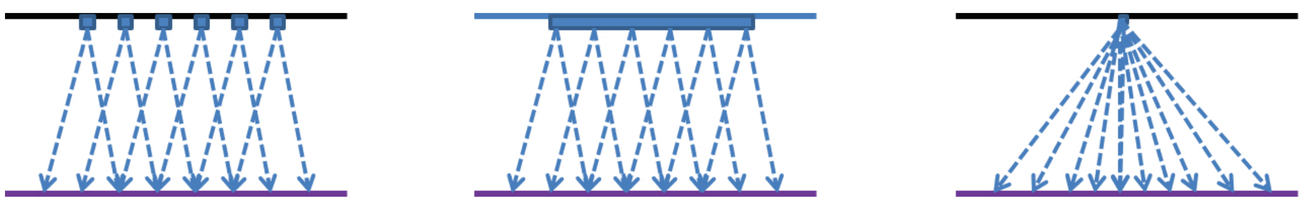
The schematic representation in **Figure A-5** highlights come key differences between the exact, FF, and NF models used throughout this annex. It is important to note that a NF model *always* requires the use of a point-source modeling assumption because of its IES file construction (x-y plane projection onto the optical center of the luminaire).

Warning: Setting up any modeling software for a point source (or for a source of the smallest physical size accepted by software that does not work with point sources, such as DIALux) is *critically important* for near-field IES files to work correctly.

Illuminance maps generated for the "ball" luminaire at 0.300 m are shown in **Figure A-6** as an example. **Table A-2** summarizes the descriptive statistics obtained from the 9x9 calculation grid. (*Note:* Throughout **Section A.2**, an ''ideal'' detector with a perfect cosine correction was assumed in the NF.) **Section A.3** provides a detailed analysis of the errors introduced by angular resolution and detector deviations from the ''ideal" response.

A.3. Effects of Angular Resolution and Angular Detector Error

Table A-3 shows a summary of the effects of vertical (θ) and horizontal (ϕ) angular resolution analyzed in "design of experiments" (DOE) fashion, calculated without incorporating the angular detector error (equivalent to an "ideal" detector), and calculated over the 9x9 grid at a 0.150-m working distance, the low limit recommended by this standard. For each beam angle, the main effect per 1° resolution in θ was comparable in magnitude to that of a 5° resolution in ϕ. *Both main effects increased*



Exact Model
100 discrete LEDs
100 FF LED IES files

FF Model
1 luminous *surface*,
1 FF Fixture IES File

NF Model
1 luminous *point*,
1 NF Fixture IES File

Figure A-5. Schematic comparison of an exact model (left), far-field model (middle), and near-field model (right). (Graphics courtesy of Emil Radkov)

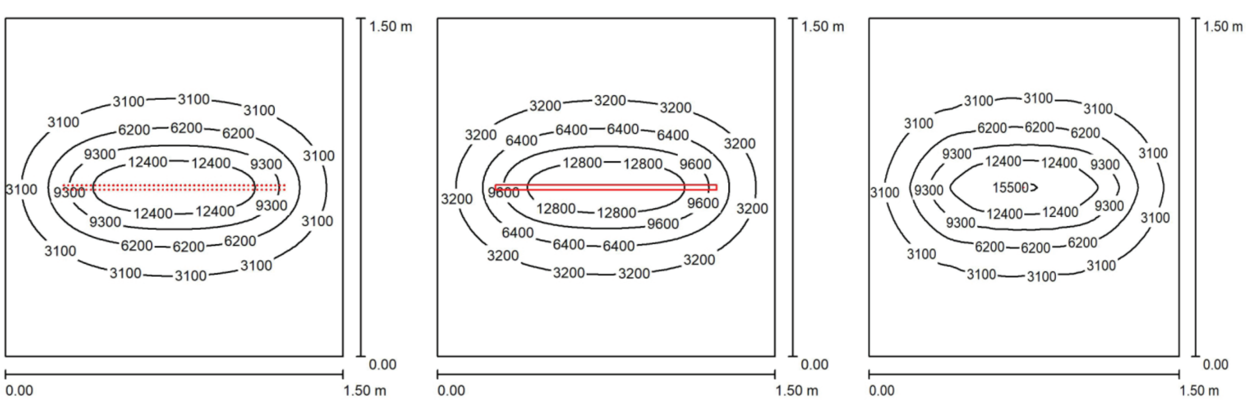


Figure A-6. Illuminance maps for a 10-klm "ball" luminaire at 0.300 m using an exact model (left), a far-field model (middle), and a near-field model (right), and assuming ''ideal'' detector response in all cases. (Graphics courtesy of Emil Radkov)

Table A-2. Comparison of Descriptive Statistics for the Three "Ball" Cases Shown in Figure A-6

Statistic	Exact Model	FF	FF/Exact	NF	NF/Exact
Minimum	374	372	0.995	374	1.000
Average	3,804	3,811	1.002	3,810	1.001
Maximum	15,592	15,848	1.016	15,592	1.000
RMSE	N.A.	N.A.	1.3%	N.A.	2.6%

sharply as the beam angle decreased in size toward a value of 60°.

Depending on its magnitude, the angular detector error $F(\theta,\phi)$ can exhibit anywhere from negligible to major effects on the lighting models using near-field photometry files. To keep such effects within bounds, this method establishes both lower and upper limits to allow for that error. Those limits are shown graphically in **Figure A-7** as staircase-shaped lines, along with the experimentally measured errors for a high-quality detector (plotted as black markers), using the procedure described in **Section 7.2**. In addition, the same figure shows the minimum and maximum error values for 4 horizontal angles at 90° intervals, connected with broken lines as functions of the absolute value of the vertical angle. (This procedure provides the detector error range due to changes in horizontal angle and reduces the problem to evaluating the vertical angle effect on detector error, as shown in Step 6 in **Section 7.2.**) It can be seen from the graph that this particular detector can be used for vertical angles up to 80°, corresponding to application distances down to 0.300 m, per the formula given in **Section 7.2**.

The most straightforward incorporation of the angular detector error into the optical models would be through the response function for the calculation grid points. That is impossible to do at the user level in DIALux (where a perfect cosine calculation point response is "hard-wired" into the software). A workaround approach is possible, however.

Table A-3. Effects of Vertical (θ) and Horizontal (ϕ) Angular Resolution for an ''Ideal'' Detector

Uncoded Angles (Degrees)		RMSE = $A \cdot \theta_{coded} + B \cdot \phi_{coded} + C \cdot \theta_{coded} \cdot \phi_{coded} + D$				
		Fixture	**Tear**	**Egg**	**Ball**	**Tomato**
		Beam A	**60°**	**90°**	**120°**	**150°**
θ	**ϕ**	**RMSE Angle Combination**	**RMSE vs. Hi-Res DIALux Model**			
1	**3**	θ_{-1}, ϕ_{-1}	3.2%	0.8%	0.3%	0.2%
4	**3**	θ_{+1}, ϕ_{-1}	48.2%	10.4%	4.8%	3.2%
1	**18**	θ_{-1}, ϕ_{+1}	33.8%	9.7%	5.7%	4.1%
4	**18**	θ_{+1}, ϕ_{+1}	87.5%	15.9%	7.5%	5.0%
A = Main θ Effect			49.3%	7.9%	3.1%	1.9%
B = Main ϕ Effect			34.9%	7.2%	4.0%	2.8%
C = θ,ϕ Interaction			4.4%	-1.7%	-1.4%	-1.0%
D = Mean RMSE for the center of the study domain at θ = 2.5°, ϕ = 10.5°			43.2%	9.2%	4.6%	3.1%
		Fixture	**Tear**	**Egg**	**Ball**	**Tomato**
		Beam	**60°**	**90°**	**120°**	**150°**
		A, per 1° θ	16.4%	2.6%	1.0%	0.6%
		B, per 5° ϕ	11.6%	2.4%	1.3%	0.9%
		C, per 5°^2	0.5%	-0.2%	-0.2%	-0.1%

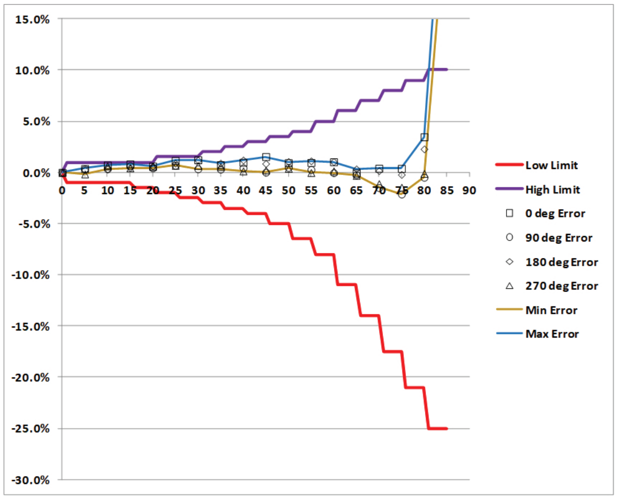


Figure A-7. Low and high error limits adopted for the detectors used in this standard and errors measured experimentally for a real detector, using the procedure in Section 7.2. (Graphic courtesy of Emil Radkov)

A small detector element with a perfect cosine response is assumed, aligned to the vertical direction (functionally equivalent to a "horizontal" calculation point in DIALux) and receiving a ray of light at a vertical angle θ from a randomly positioned LED in the luminaire, per **Figure A-8.** That ray of light has left the LED also at a vertical angle θ, since the exit vertical angle from the LED and the entry vertical angle into the detector are *alternate interior angles*, formed by the straight line connecting the detector and LED and the vertical lines passing through each of them. (As a reminder, in geometry, alternate interior angles are congruent.)

Therefore, for the purpose of incorporating the effect of the detector in model simulations, the workaround solution is to first apply the detector error $F(\theta,\phi)$ *at each*

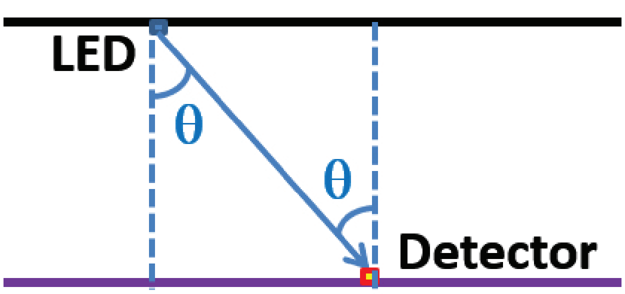


Figure A-8. Illustration of the approach used to incorporate vertical angular detector error into the optical models used here. (Graphic courtesy of Emil Radkov)

vertical angle to the IES file of each LED and then use those modified IES files to build exact optical models and collect data from the calculation points in DIALux.

The next step of this analysis treats the lower and upper detector-error limits mathematically as actual detectors and evaluates the associated errors after incorporation of the corresponding errors into lighting models, as described in **Sections A.2** and **A.3**. Then combinations of vertical and horizontal angular resolutions are selected such that after correction for total flux, the associated RMSE values remained within 15% and the average values remain within 3% of those from the exact model at an application distance of 0.150 m using an "ideal" detector—for both the lower and upper error limits. In addition, the tolerances in FF determination of beam angles are included in the range for the upper error limits as double values, to account for the 2-sided definition of the beam angle. For example, the values of 5° in θ and 22.5° in ϕ were found to satisfy the aforementioned criteria for beams with angles of 120° and above. Taking double the 5° θ tolerance into account for beam angle far-field determination, the acceptable beam angle range was adjusted to 110° and above. The other angle ranges limited by 84° and 58° quoted in the document were obtained in a similar fashion. Given the relatively large RMS and average error allowances used in the estimation of the minimal angular resolution requirements listed in this standard (as well as the possibility for different outcomes with other fixture geometries), the users of this standard are advised to select finer angular resolutions whenever feasible.

A.4. Data Extension at High Angles

Data collection for the generation of NF IES files requires progressively larger scanning radii, going out to infinity at a vertical angle of 90°. This is because the scanning radius is a tangent function of the vertical angle, θ, which becomes infinitely large as θ approaches 90°. Since the track length of any real NF light detector is finite (referred to below as r_{lim}), it is necessary to find a way to extend the NF dataset for any vertical angles requiring data collection beyond that length. The Technical Group examined two possible extensions at high vertical angles, summarized here.

The first such extension was provided in LM-70-2000 (currently deprecated by the IES) and requires repositioning the detector upon reaching $r_{\lim}$ at vertical z distances from the luminaire progressively shorter than the working one, to achieve the necessary θ values there. This amounts to adding a new testing procedure, which requires moving the detector vertically to $z = r_{\lim}\cdot\cot\ \theta$, instead of *horizontally to* $r_{(z,\theta)} = z\cdot\tan\ \theta$, as in the typical NF data collection mode—with the corresponding increase in labor, measuring time, and data processing complexity. In addition, this extension was found to distort the polar plots of the IES files at high angles, particularly for the luminaires with beam angles greater than 120°.

The graphical example in **Figure A-9** shows side-by-side IES polar plots for the "tomato" luminaire, all calculated at a working distance of 0.150 m, with detector error included, first in an exact model out to 88° (with the values beyond that zeroed out) entirely in NF (left), then using the first extension discussed above after 78° (center), and finally with the second extension discussed below after 78° (right). It is important to note the "bump" at high angles in the middle graph (inside the dashed-line circle), absent in both other graphs.

The second such extension uses far-field data at the high θ values for the same purpose. (Note that FF data collection is *required* for the lumen correction of the NF data, per this standard.) This extension provides polar plots closer to the non-extended model, as illustrated in the preceding example. This is because the progression to ever larger scanning radii necessary at the higher vertical angles implies satisfying the FF requirement at some point, meaning that transition to FF behavior amounts to a basic physical expectation. This extension was selected by the Technical Group for implementation into the standard.

For all practical purposes, experimental FF extension data may be assumed to be unaffected by the angular detector errors discussed in **Section A.3**, since they are collected with the detector oriented toward the optical center of the luminaire, which results in an associated angular detector error close to 0. For this reason, any FF extensions are excluded from the correction applied subsequently to the NF part of the dataset, as directed in **Section 9.3**.

The available detector track length in any given NF measurement system effectively determines at what vertical angle the FF data extension has to start. This standard requires that this length be least 1.75 m, forming a circle of diameter 3.50 m around the projection of the optical center of the DUT on the working plane. This ensures NF data collection out to vertical angles of at least 85° for the working distance of 0.15 m, from where the radius of 1.700 m comes into the minimal distance formula in **Section 7.2**, with 5 cm of design margin added for the minimal detector track length required in **Section 4.7.1.1**. The transition to FF data can appear disjointed if done at lower angles. It should also be noted that any FF data extensions provide optical model data for irradiance maps under a

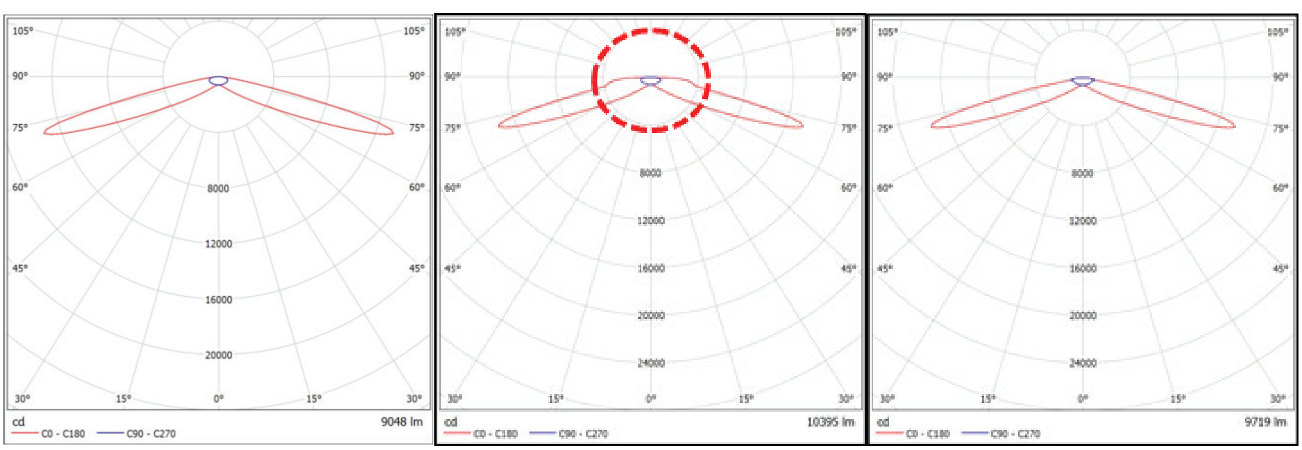


Figure A-9. Polar plots of IES files for a 10-klm "tomato" luminaire at 0.150 m, incorporating detector error in an exact model (left), extended per LM-70-2000 (middle) and with far-field values (right). (Graphics courtesy of Emil Radkov) single luminaire *only beyond a circle of the same 1.75-m radius*, which encompasses the outline of virtually any existing luminaire.

A.5. Vertical Interpolation

Interpolation of near-field data in the vertical (z) direction is of interest, since such data are valid only for the fixed working distance at which they are collected. The use of exact models also allows for independent evaluation of the associated errors, as summarized here.

Model data for at each of the five chosen beam angles at a given working distance were interpolated using two planes at shorter and longer z distances (e.g., the plane at z = 30 cm was interpolated using the planes at z = 15 cm and z = 45 cm), assuming ideal detector behavior with no angular errors.

Illuminance and *z* values were taken either directly from the models or after various functional transforms and combinations thereof (logarithmic, exponential, power, or parametric power with independent optimization of the exponent). The following general interpolation formula was used:

$$Y_i = Y_1 + (X_i - X_1)(Y_2 - Y_1)/(X_2 - X_1) ,$$

where:

Y = illuminance values after transform

X = vertical distance values after transform

i = index referring to the plane being interpolated; the indices 1 and 2 refer to the two model planes used for interpolation

Next, the interpolated values were converted back to E (illuminance) and z values by applying the inverse transforms to each part, and the results were compared against the exact model for the intermediate z value on the coarse calculation grids introduced in **Section A.1**. **Table A-4** provides a representative example interpolating the calculation grid for the "ball" beam, with power functions giving relatively reduced errors.

It can be seen that the RMSE value introduced even by this "optimized" interpolation was still above 10%. However, the RMSE value introduced by using a FF model instead of the NF exact model was about an order of magnitude lower (see **Table A-5**). This was the case across the board. As the beam angle decreased from 150° to 60°, errors in the far-field models at 30 cm increased 3-fold (but interpolation errors increased 10-fold). The RMSE for transformed interpolation stayed *above* 10% at 0.300 m, even in the best "tomato" case. By contrast, the RMSE for far-field DIALux models stayed *below* 4% at 0.300 m, even in the worst "tear" case.

Individual 3-point quadratic fits were also attempted, with 3 individual fit parameters *for every grid point* (with and without the use of transforms), using exact models at 0.200, 0.600 and 1.000 m as the basis of the fits. No significant further improvement was seen in the interpolation accuracy at the intermediate distances.

Table A-4. Example of Vertical Interpolation Error (%) for the ''Ball'' Beam at 0.300 m

	Distance, m								
Distance, m	**0.083**	**0.250**	**0.417**	**0.583**	**0.750**	**0.917**	**1.083**	**1.250**	**1.417**
1.417	15.5	12.6	10.8	9.9	9.7	9.9	10.8	12.6	15.5
1.250	6.7	2.1	-0.2	-1.1	-1.2	-1.1	-0.2	2.1	6.7
1.083	-4.0	-10.5	-13.4	-13.9	-13.9	-13.9	-13.4	-10.5	-4.0
0.917	-12.8	-16.4	-18.4	-19.2	-19.1	-19.2	-18.4	-16.4	-12.8
0.750	-14.3	-1.0	-0.1	-2.9	-3.1	-2.9	-0.1	-1.0	-14.3
0.583	-12.8	-16.4	-18.4	-19.2	-19.1	-19.2	-18.4	-16.4	-12.8
0.417	-4.0	-10.5	-13.4	-13.9	-13.9	-13.9	-13.4	-10.5	-4.0
0.250	6.7	2.1	-0.2	-1.1	-1.2	-1.1	-0.2	2.1	6.7
0.083	15.5	12.6	10.8	9.9	9.7	9.9	10.8	12.6	15.5
RMSE = 11.6%									

Table notes: Interpolating between 15 cm and 45 cm. E^m, z^p: m = 0.5; p = -0.5.

Table A-5. Far-Field DIALux Model Error (%) for the ''Ball'' Beam at 0.300 m

Distance, m	Distance, m								
	0.083	**0.250**	**0.417**	**0.583**	**0.750**	**0.917**	**1.083**	**1.250**	**1.417**
1.417	-0.5	-0.2	0.2	0.5	0.5	0.5	0.2	-0.2	-0.5
1.250	-1.1	-0.6	0.1	0.6	0.8	0.6	0.1	-0.6	-1.1
1.083	-2.0	-1.0	0.2	0.9	1.0	0.9	0.2	-1.0	-2.0
0.917	-3.2	-1.6	0.4	1.2	1.4	1.2	0.4	-1.6	-3.2
0.750	-3.9	-1.9	0.6	1.5	1.6	1.5	0.6	-1.9	-3.9
0.583	-3.2	-1.6	0.4	1.2	1.4	1.2	0.4	-1.6	-3.2
0.417	-2.0	-1.0	0.2	0.9	1.0	0.9	0.2	-1.0	-2.0
0.250	-1.1	-0.6	0.1	0.6	0.8	0.6	0.1	-0.6	-1.1
0.083	-0.5	-0.2	0.2	0.5	0.5	0.5	0.2	-0.2	-0.5
RMSE = 1.3%									

Accordingly, the best practical recommendation to be made at this time regarding working planes between those used in NF data collection is *not to use vertical interpolation but to construct FF models for such z values instead.*

A.6 Light Beams with Two Characteristic Angles

Some FF DUT light beams, such as the theoretical "bumblebee" type discussed previously, have more than one characteristic angle. This not only complicates their beam angle definition but also raises questions as to which (if any) characteristic angle dominates the overall beam "behavior" and how that in turn affects the minimum angular resolution necessary for data acquisition. The following analysis addresses those questions with the help of an extreme example of a beam having both a very small (15°) and a very large (165°) characteristic angle (see **Figure A-10**).

One way to look for answers is by checking how this new beam shape compares to the first 4 theoretical beam shapes in any *clear* trends observed for them. Using such trends, an *"effective beam angle" could be assigned, strictly for the purposes of this analysis,* to any other beam according to the nearest theoretical beam angle(s), within approximately 30°.

Important to recall: Each of the first four theoretical beam shapes ("tear," "egg," "ball," "tomato") introduced in **Section A.1** had only one characteristic angle, so there is no ambiguity in the beam angle definition in that regard.

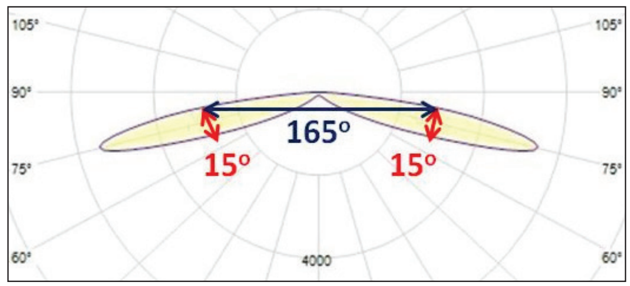


Figure A-10. An extreme beam example, with both very small and very large characteristic angles. (Graphic courtesy of Emil Radkov)

The new beam shape introduced in **Figure A-10** will be referred to as "narrow batwing" and abbreviated as NBW later in this section. It was scaled to the same overall luminous flux (10 klm at FF fixture level) as in all preceding examples and used in the same luminaire configuration in the DIALux models, for full consistency of all comparisons.

The first trends examined were those for zonal lumens, presented in **Figure A-11** at individual LED level after correction for a real detector, following the low vertical error limit, according to **Section A.3**. (The zonal lumens trend for the ideal detector is virtually identical.)

Based on the trend in angles of zones containing the most lumens (highlighted in each part of **Figure A-11**), the *effective angle* for ''narrow batwing'' (NBW) is greater than 150°. It is important to note that the maximum lumens zone in NBW is *two* ranges higher than that for "tomato" and contains *about half* of the total lumens.

An additional trend (observable only with real detectors since with the ideal one, all LED total lumens here would be equal to 100 lm) can be seen for the total lumens at the bottom of each part of **Figure A-11**. That trend also supports an effective NBW beam angle of greater than 150°.

The next trend is the one in NF RMSE values using an ideal detector for the 9x9 calculation grids (introduced in **Section A.1** and mentioned in **Sections A.2**, **A.3**, and **A.5**), listed in **Figure A-12** for a working distance of 0.150 m at maximum vertical and horizontal angle resolution.

According to this trend, the effective angle for NBW is between 120° and 150° for the ideal detector. (This may be due to a "pull" by the smaller characteristic angle.)

The effects of angular resolution presented in **Section A.3** are not as clear in the NBW case (with the exception of a coefficient *B* of 0.2% in its DOE model for ideal detector, supporting an effective beam angle greater than 150°) because of the non-monotonous behavior of the remaining DOE coefficients for the other four beams, compared to those of NBW (*A* = 3.3%, *C* = -0.3%, *D* = 2.2%). The latter values are consistent with an effective beam angle greater than 150° for NBW but also allow the alternative of a smaller such angle, as long as it is greater than 60°.

Regardless of some indications of possible "pull" from the 15° characteristic angle, *none of the trends examined*

Tear

ZONE	LUMENS	% TOTAL
0-10	8.7	8.9%
10-20	22.1	22.7%
20-30	26.1	26.8%
30-40	21.3	21.9%
40-50	12.6	12.9%
50-60	5.1	5.3%
60-70	1.2	1.3%
70-80	0.1	0.1%
80-90	0.002	0.0%
0-90	**97.2**	**100%**

Egg

ZONE	LUMENS	% TOTAL
0-10	4.4	4.7%
10-20	12.4	13.0%
20-30	17.6	18.5%
30-40	19.4	20.3%
40-50	17.6	18.5%
50-60	13.1	13.7%
60-70	7.5	7.9%
70-80	2.8	3.0%
80-90	0.4	0.4%
0-90	**95.2**	**100%**

Ball

ZONE	LUMENS	% TOTAL
0-10	3.0	3.2%
10-20	8.6	9.2%
20-30	13.0	13.9%
30-40	15.8	16.9%
40-50	16.6	17.8%
50-60	15.2	16.2%
60-70	11.7	12.5%
70-80	7.1	7.6%
80-90	2.4	2.6%
0-90	**93.4**	**100%**

Tomato

ZONE	LUMENS	% TOTAL
0-10	2.2	2.4%
10-20	6.6	7.1%
20-30	10.3	11.2%
30-40	13.1	14.3%
40-50	14.8	16.1%
50-60	15.0	16.4%
60-70	13.5	14.7%
70-80	10.4	11.3%
80-90	5.9	6.5%
0-90	**91.8**	**100%**

Bumblebee

ZONE	LUMENS	% TOTAL
0-10	0.5	0.6%
10-20	3.4	3.7%
20-30	8.3	9.1%
30-40	13.6	14.8%
40-50	17.6	19.2%
50-60	18.6	20.3%
60-70	15.8	17.3%
70-80	10.2	11.1%
80-90	3.6	3.9%
0-90	**91.6**	**100%**

Narrow Batwing

ZONE	LUMENS	% TOTAL
0-10	0.1	0.1%
10-20	0.3	0.3%
20-30	0.5	0.6%
30-40	0.9	1.1%
40-50	2.2	2.7%
50-60	6.3	7.5%
60-70	18.5	22.2%
70-80	41.1	49.4%
80-90	13.4	16.1%
0-90	**83.3**	**100%**

Figure A-11. Zonal-lumen and total-lumen trends for LEDs with different beam profiles, including vertical-angle detector error. (Image courtesy of Emil Radkov)

Descriptive Statistics Summary (Tear):

	Exact Model	NF-Ideal Detector	
Min	0.42	0.42	1.000
Avg	5148	5143	0.999
Max	61532	61532	1.000
RMSE	N/A (benchmark)		3.24%

Descriptive Statistics Summary (Egg):

	Exact Model	NF-Ideal Detector	
Min	31	31	1.000
Avg	4591	4589	1.000
Max	41954	41954	1.000
RMSE	N/A (benchmark)		0.81%

Descriptive Statistics Summary (Ball):

	Exact Model	NF-Ideal Detector	
Min	113	113	1.000
Avg	4345	4343	1.000
Max	32768	32768	1.000
RMSE	N/A (benchmark)		0.34%

Descriptive Statistics Summary (Tomato):

	Exact Model	NF-Ideal Detector	
Min	201	201	1.000
Avg	4097	4096	1.000
Max	27090	27090	1.000
RMSE	N/A (benchmark)		0.24%

Descriptive Statistics Summary (Bumblebee):

	Exact Model	NF-Ideal Detector	
Min	167	167	1.000
Avg	3974	3973	1.000
Max	21039	21039	1.000
RMSE	N/A (benchmark)		0.17%

Descriptive Statistics Summary (Narrow Batwing):

	Exact Model	NF-Ideal Detector	
Min	852	852	1.000
Avg	3357	3354	0.999
Max	5951	5954	1.001
RMSE	N/A (benchmark)		0.29%

Figure A-12. Trend in NF RMSE values using an ideal detector for the 9x9 calculation grids and z = 0.150 m.
(Image courtesy of Emil Radkov)

in this work was consistent with an effective NBW beam angle below 60°.

In conclusion, the *larger* characteristic angle prevails across the board and is the most relevant one to use for the NBW beam definition (and, for the same reason, for the "bumblebee"—as can be seen by inspection of its data, shown throughout this section). This recommendation is also supported by the analysis of the FF models (not shown), which gave the smallest RMSE value for NBW among all cases, even at the coarsest vertical angle resolution of 5°.

Annex B – Sample Computer LM-63 and TM-33 Files

This annex provides an example of data processing for a 10-kilolumen (klm) linear luminaire with a Lambertian beam in far-field conditions. **Table B-1** shows how raw illuminance data from a near-field measurement are transformed, corrected for luminous flux, and merged with far-field data of higher symmetry as part of the generation of a distance-specific photometric file.

Table B-1. Example of Raw Illuminance NF Data Transformation, Correction, and Merging With FF Data.

Illuminance values on the application plane (z=0.300 m), lx

φ,°↓ θ,°:	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90
0.0	15204	15193	15164	15112	15030	14908	14730	14464	14051	13399	12284	10397	7295	3654	1165	245	33	N/A	N/A
22.5	15204	15168	15058	14874	14595	14209	13697	13009	12095	10909	9316	7297	4830	2472	880	211	31	N/A	N/A
45.0	15204	15108	14809	14324	13623	12716	11634	10354	8900	7331	5663	4035	2511	1320	539	156	28	N/A	N/A
67.5	15204	15046	14567	13800	12751	11473	10053	8508	6919	5382	3925	2674	1632	872	383	125	25	N/A	N/A
90.0	15204	15022	14468	13596	12418	11020	9504	7903	6308	4815	3453	2326	1407	758	339	115	24	N/A	N/A

θ,°:	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90
Farfield cd	3183	3171	3135	3075	2991	2885	2757	2607	2438	2251	2046	1826	1592	1345	1089	824	553	277	0

Transformed data (with $z^2\cos^{-3}\theta$), merged with FF at 85 deg

φ,°↓ θ,°:	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85 (From FF)	90
0.0	1368	1383	1429	1509	1630	1802	2041	2368	2813	3411	4163	4959	5252	4357	2621	1272	567	277	0
22.5	1368	1381	1419	1485	1583	1718	1898	2130	2422	2777	3157	3480	3478	2947	1980	1095	533	277	0
45.0	1368	1375	1395	1430	1478	1537	1612	1695	1782	1866	1919	1924	1808	1574	1212	810	481	277	0
67.5	1368	1370	1373	1378	1383	1387	1393	1393	1385	1370	1330	1275	1175	1040	862	649	430	277	0
90.0	1368	1368	1363	1358	1347	1332	1317	1294	1263	1226	1170	1109	1013	904	763	597	413	277	0

NF lumen-corrected data, cd (ready for NF IES file creation)*

φ,°↓ θ,°:	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85 (From FF)	90
0.0	1419	1434	1482	1565	1691	1869	2117	2456	2918	3538	4317	5143	5447	4519	2718	1319	588	277	0
22.5	1419	1432	1472	1541	1642	1782	1968	2209	2511	2880	3274	3610	3607	3057	2053	1136	553	277	0
45.0	1419	1426	1447	1484	1532	1594	1672	1758	1848	1935	1990	1996	1875	1632	1258	840	499	277	0
67.5	1419	1421	1424	1429	1434	1439	1445	1445	1437	1421	1379	1323	1219	1078	894	673	446	277	0
90.0	1419	1418	1414	1408	1397	1382	1366	1342	1310	1271	1214	1151	1051	937	791	619	428	277	0

*This example used numerical integration in the NF and FF following the rectangle rule. The corrected values from this step may become slightly different with other integration formulas.

B.1 Example of Far-Field IES File with Rotational Symmetry

```
IES:LM-63-2019
[TEST] Example in FF
[TESTLAB] IES TPC
[ISSUEDATE] 11/03/2020 9:00:00 AM
[MANUFAC] Company A
[LUMINAIRE] 10 klm Lambertian Linear
[LAMP] 100 Square 3030 LED
TILT=NONE
1 -1 1 19 1 1 2 0.050 1.000 0.050
1.000 1 100
0    5     10    15    20    25    30    35    40    45    50    55    60    65
70   75    80    85    90
0
3183 3171  3135  3075  2991  2885  2757  2607  2438  2251  2046  1826  1592  1345
1089 824   553   277   0
```

B.2 Example of Near-Field IES File of the Same Luminaire with 2-Fold Mirror Symmetry (Incorporating the Maximum Allowed Lower-Limit Detector Error and Lumen-Corrected in NF to 80°) in LM-63 Format

```
IES:LM-63-2019
[TEST] Example at 0.300 m NF
[NEARFIELD] 0.300,0,0
[MORE] THIS FILE MAY NOT BE USED FOR ANY APPLICATION DISTANCES OTHER THAN THE
ONE LISTED ABOVE
[TESTLAB] IES TPC
[ISSUEDATE] 11/03/2020 10:00:00 AM
[MANUFAC] Company A
[LUMINAIRE] 10 klm Lambertian Linear
[LAMP] 100 Square 3030 LED
[OTHER] THE ACTUAL LUMINOUS DIMENSIONS MAY NOT BE USED IN NEARFIELD LIGHTING
CALCULATIONS
[OTHER] LUMINAIRE PHYSICAL DIMENSIONS: 0.050, 1.000, 0.050 m
[OTHER] THIS IS A RECTANGULAR LUMINAIRE
[OTHER] MINIMAL APPLICATION DISTANCE ALLOWED BY THE DETECTOR ERROR: 0.300 m
[OTHER] VERTICAL ANGLE WHERE NEARFIELD TRANSITIONS TO FARFIELD: 85 deg
[OTHER] ASSUMED SYMMETRY: 2-FOLD MIRROR SYMMETRY VS. C0-C180 and C90-C270
TILT=NONE
1 -1 1 19 5 1 2 0.001 0.001 0.001
1.000 1 100
0    5     10    15    20    25    30    35    40    45    50    55    60    65
70   75    80    85    90
0.0  22.5  45.0  67.5  90.0
1463 1479  1528  1614  1743  1927  2182  2532  3008  3647  4451  5302  5616  4658
2802 1360  606   277   0
1463 1476  1517  1588  1693  1837  2029  2277  2589  2969  3375  3721  3718  3151
2116 1171  570   277   0
```

```
1463 1470   1492   1529   1580   1644   1724   1813   1905   1995   2052   2058   1933   1683
1296 866    515    277    0
1463 1464   1468   1473   1479   1483   1489   1489   1481   1465   1422   1364   1256   1112
921  694    459    277    0
1463 1462   1458   1452   1440   1424   1408   1384   1350   1310   1251   1186   1083   966
815  638    441    277    0
```

B.3 Same Example, in TM-33 Format


```
<?xml version = „1.0“?>
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  <!-- Original file: 10klmBall30cmRD19x5LDL-LMv1.ies -->
  <Version>1.0</Version>
  <Header>
    <Manufacturer>Company A</Manufacturer>
    <CatalogNumber>--MISSING--</CatalogNumber>
    <Description>10 klm Lambertian Linear</Description>
    <Laboratory>IES TPC</Laboratory>
    <ReportNumber>Example at 0.300 m NF</ReportNumber>
    <ReportDate>1970-01-01</ReportDate>
    <Comment>IES:LM-63-2019</Comment>
    <Comment>[TEST] Example at 0.300 m NF</Comment>
    <Comment>[NEARFIELD] 0.300,0,0</Comment>
    <Comment>[MORE] THIS FILE MAY NOT BE USED FOR ANY APPLICATION DISTANCES
OTHER THAN THE ONE LISTED ABOVE</Comment>
    <Comment>[TESTLAB] IES TPC</Comment>
    <Comment>[ISSUEDATE] 11\/03\/2020 10:00:00 AM</Comment>
    <Comment>[MANUFAC] Company A</Comment>
    <Comment>[LUMINAIRE] 10 klm Lambertian Linear</Comment>
    <Comment>[LAMP] 100 Square 3030 LED</Comment>
    <Comment>[OTHER] THE ACTUAL LUMINOUS DIMENSIONS MAY NOT BE USED IN NEARFIELD
LIGHTING CALCULATIONS</Comment>
    <Comment>[OTHER] LUMINAIRE PHYSICAL DIMENSIONS: 0.050, 1.000, 0.050 m</
Comment>
    <Comment>[OTHER] THIS IS A RECTANGULAR LUMINAIRE</Comment>
    <Comment>[OTHER] MINIMAL APPLICATION DISTANCE ALLOWED BY THE DETECTOR ERROR:
0.300 m</Comment>
    <Comment>[OTHER] VERTICAL ANGLE WHERE NEARFIELD TRANSITIONS TO FARFIELD: 85
deg</Comment>
<Comment>[OTHER] ASSUMED SYMMETRY: 2-FOLD MIRROR SYMMETRY VS. C0-C180 and C90-
C270</Comment>
  </Header>
  <Luminaire>
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      <Length>0.0</Length>
      <Width>0.0</Width>
      <Height>0.0</Height>
    </Dimensions>
```

```
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  </Luminaire>
  <Equipment>
    <Gonioradiometer>
      <Type>IES _ C</Type>
    </Gonioradiometer>
  </Equipment>
  <Emitter>
    <Quantity>1</Quantity>
    <Description>100 Square 3030 LED</Description>
    <CatalogNumber>--MISSING--</CatalogNumber>
    <RatedLumens>-1.0</RatedLumens>
    <InputWattage>1.0</InputWattage>
    <BallastFactor>1.0</BallastFactor>
    <LuminousData>
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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

a) Proposed change accepted for public review without modification
b) Proposed change accepted for public review with modification
c) Proposed change accepted for further study
d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: ______________________________
   Affiliation: ______________________________
   Address: ______________________________
   City: ______________ State: ________ Zip: ________ Country: ______________
   Telephone: ______________________________
   Fax: ______________________________
   E-mail: ______________________________

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: ______________________________ Date: ______________

2. Title of publications and year published ______________________________
3. Clause (section), sub-clause or paragraph number; and page number: ______________________________
4. My proposal (check one):
   [ ] Change to read as follows
   [ ] Delete and substitute as follows
   [ ] Add new text as follows
   [ ] Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

____
____
____

6. Reason and substantiation:

____
____
____

Select as applicable:
[ ] Additional pages are attached. Number of additional pages:
[ ] Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:

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